ECOLOGICAL SUCCESSION AND BIOMES

SUCCESSION DECISION TREE → PRIMARY SUCCESSION RAIL → SECONDARY SUCCESSION RAIL → SERE VOCABULARY MAP → HYDRARCH AND XERARCH COMPARISON → MECHANISM FORK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 SUCCESSION DECISION TREE

SUCCESSION DECISION TREE | NEWLY EXPOSED SUBSTRATE OR DISTURBED FORMER COMMUNITY | NO DEVELOPED SOIL | PRIMARY SUCCESSION PATHWAY | SOIL AND BIOLOGICAL LEGACIES REMAIN | SECONDARY SUCCESSION PATHWAY | AUTOGENIC COMMUNITY CHANGE PLUS ALLOGENIC EXTERNAL FORCING | CONTINGENT TRAJECTORY, NOT ONE GUARANTEED ENDPOINT

START → newly exposed substrate or disturbed former community

NO DEVELOPED SOIL → primary succession pathway → SOIL AND BIOLOGICAL LEGACIES REMAIN → secondary succession pathway → DRIVERS → autogenic community change plus allogenic external forcing

SYSTEM INPUT / STARTING CONDITION
- START → newly exposed substrate or disturbed former community
- NO DEVELOPED SOIL → primary succession pathway

ECOLOGICAL PROCESS / TRANSFER
- SOIL AND BIOLOGICAL LEGACIES REMAIN → secondary succession pathway
- DRIVERS → autogenic community change plus allogenic external forcing

OUTCOME / FEEDBACK / LIMIT
- OUTCOME → contingent trajectory, not one guaranteed endpoint
- START → newly exposed substrate or disturbed former community

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → contingent trajectory, not one guaranteed endpoint.

01

STAGE 01 PRIMARY SUCCESSION RAIL

PRIMARY SUCCESSION RAIL | BARE OR NEW SUBSTRATE | PIONEER ESTABLISHMENT | WEATHERING AND ORGANIC INPUT | INITIAL SOIL DEVELOPMENT | HERBS OR OTHER COLONISTS | MORE COVER AND RESOURCE CAPTURE | LATER COMMUNITIES

BARE OR NEW SUBSTRATE → pioneer establishment → WEATHERING AND ORGANIC INPUT → initial soil development → HERBS OR OTHER COLONISTS → more cover and resource capture → LATER COMMUNITIES → increasing structural complexity where conditions permit → LIMIT → no universal species list or completion time

SYSTEM INPUT / STARTING CONDITION
- BARE OR NEW SUBSTRATE → pioneer establishment
- WEATHERING AND ORGANIC INPUT → initial soil development

ECOLOGICAL PROCESS / TRANSFER
- HERBS OR OTHER COLONISTS → more cover and resource capture
- LATER COMMUNITIES → increasing structural complexity where conditions permit

OUTCOME / FEEDBACK / LIMIT
- LIMIT → no universal species list or completion time
- BARE OR NEW SUBSTRATE → pioneer establishment

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIMIT → no universal species list or completion time.

02

STAGE 02 SECONDARY SUCCESSION RAIL

SECONDARY SUCCESSION RAIL | PRIOR VEGETATION REDUCED OR REMOVED | SOIL, ROOTS, SEEDS OR MICROBES REMAIN | BIOLOGICAL LEGACY | RAPID COLONISERS | EARLY RECOVERY | COMPETITION AND RECRUITMENT | LATER COMMUNITY CHANGE

DISTURBANCE → prior vegetation reduced or removed → SOIL, ROOTS, SEEDS OR MICROBES REMAIN → biological legacy → RAPID COLONISERS → early recovery → COMPETITION AND RECRUITMENT → later community change → LIMIT → faster than primary does not mean instant or identical recovery

SYSTEM INPUT / STARTING CONDITION
- DISTURBANCE → prior vegetation reduced or removed
- SOIL, ROOTS, SEEDS OR MICROBES REMAIN → biological legacy

ECOLOGICAL PROCESS / TRANSFER
- RAPID COLONISERS → early recovery
- COMPETITION AND RECRUITMENT → later community change

OUTCOME / FEEDBACK / LIMIT
- LIMIT → faster than primary does not mean instant or identical recovery
- DISTURBANCE → prior vegetation reduced or removed

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIMIT → faster than primary does not mean instant or identical recovery.

03

STAGE 03 SERE VOCABULARY MAP

SERE VOCABULARY MAP | COMPLETE SEQUENCE AT ONE SITE | SERAL STAGE | ONE COMMUNITY WITHIN THE SEQUENCE | ROCK PSAMMOSERE | SALINE HYDROSERE | FRESH WATER | DRY SUBSTRATE; NAMES IDENTIFY STARTS, NOT FIXED ENDS

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
SERE → complete sequence at one site	LITHOSERE → rock PSAMMOSERE → sand	XEROSERE → dry substrate; names identify starts, not fixed ends
SERAL STAGE → one community within the sequence	HALOSERE → saline HYDROSERE → fresh water	SERE → complete sequence at one site

SYSTEM INPUT / STARTING CONDITION
- DISTURBANCE → prior vegetation reduced or removed
- SOIL, ROOTS, SEEDS OR MICROBES REMAIN → biological legacy

ECOLOGICAL PROCESS / TRANSFER
- RAPID COLONISERS → early recovery
- COMPETITION AND RECRUITMENT → later community change

OUTCOME / FEEDBACK / LIMIT
- LIMIT → faster than primary does not mean instant or identical recovery
- DISTURBANCE → prior vegetation reduced or removed

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIMIT → faster than primary does not mean instant or identical recovery.

03

STAGE 03 SERE VOCABULARY MAP

SERE VOCABULARY MAP · COMPLETE SEQUENCE AT ONE SITE · SERAL STAGE · ONE COMMUNITY WITHIN THE SEQUENCE · ROCK PSAMMOSERE · SALINE HYDROSERE · FRESH WATER · DRY SUBSTRATE; NAMES IDENTIFY STARTS, NOT FIXED ENDS

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
SERE → complete sequence at one site	LITHOSERE → rock PSAMMOSERE → sand	XEROSERE → dry substrate; names identify starts, not fixed ends
SERAL STAGE → one community within the sequence	HALOSERE → saline HYDROSERE → fresh water	SERE → complete sequence at one site

ECOLOGICAL UNIT / PARAMETER
- SERE → complete sequence at one site
- SERAL STAGE → one community within the sequence

DECISIVE DISTINCTION
- LITHOSERE → rock PSAMMOSERE → sand
- HALOSERE → saline HYDROSERE → fresh water

SCALE / STATUS / EXCEPTION
- XEROSERE → dry substrate; names identify starts, not fixed ends
- SERE → complete sequence at one site

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: XEROSERE → dry substrate; names identify starts, not fixed ends.

04

STAGE 04 HYDRARCH AND XERARCH COMPARISON

HYDRARCH AND XERARCH COMPARISON · BEGINS IN WATER OR VERY WET SUBSTRATE · BEGINS UNDER DRY CONDITIONS · MAY TREND TOWARD MORE MESIC REGIONAL CONDITIONS · DEPENDS ON SEDIMENT, SOIL, COLONISTS AND DISTURBANCE · CONVERGENCE IS NOT ONE UNIVERSAL SEQUENCE

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
HYDRARCH → begins in water or very wet substrate	BOTH → may trend toward more mesic regional conditions	CAUTION → convergence is not one universal sequence
XERARCH → begins under dry conditions	PATH → depends on sediment, soil, colonists and disturbance	HYDRARCH → begins in water or very wet substrate

ECOLOGICAL UNIT / PARAMETER
- HYDRARCH → begins in water or very wet substrate
- XERARCH → begins under dry conditions

DECISIVE DISTINCTION
- BOTH → may trend toward more mesic regional conditions
- PATH → depends on sediment, soil, colonists and disturbance

SCALE / STATUS / EXCEPTION
- CAUTION → convergence is not one universal sequence
- HYDRARCH → begins in water or very wet substrate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAUTION → convergence is not one universal sequence.

05

STAGE 05 MECHANISM FORK

MECHANISM FORK · ORGANISMS ALTER LITTER, SOIL, SHADE OR MICROCLIMATE · FLOOD, FIRE, EROSION, DEPOSITION, CLIMATE OR LAND USE ACTS · EARLY SPECIES IMPROVE LATER ESTABLISHMENT · INHIBITION OR TOLERANCE · ALTERNATIVE REPLACEMENT MECHANISMS · IDENTIFY THE DRIVER BEFORE NAMING THE MECHANISM

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
AUTOGENIC → organisms alter litter, soil, shade or microclimate	FACILITATION → early species improve later establishment	RULE → identify the driver before naming the mechanism
ALLOGENIC → flood, fire, erosion, deposition, climate or land use acts externally	INHIBITION OR TOLERANCE → alternative replacement mechanisms	AUTOGENIC → organisms alter litter, soil, shade or microclimate

ECOLOGICAL UNIT / PARAMETER
- AUTOGENIC → organisms alter litter, soil, shade or microclimate
- ALLOGENIC → flood, fire, erosion, deposition, climate or land use acts externally

DECISIVE DISTINCTION
- FACILITATION → early species improve later establishment
- INHIBITION OR TOLERANCE → alternative replacement mechanisms

SCALE / STATUS / EXCEPTION
- RULE → identify the driver before naming the mechanism
- AUTOGENIC → organisms alter litter, soil, shade or microclimate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → identify the driver before naming the mechanism.

06

STAGE 06 CLIMAX THEORY PRESSURE TEST

CLIMAX THEORY PRESSURE TEST · CLASSICAL CLAIM · REGIONAL CLIMATE FAVOURS A RELATIVELY STABLE CLIMAX · DISTURBANCE HISTORY AND SPECIES ARRIVAL DIFFER AMONG SITES · MODERN READING · MULTIPLE PLAUSIBLE STABLE STATES CAN OCCUR · REDIRECTS OR RESETS TRAJECTORIES · CLIMATE CONSTRAINS BUT DOES NOT UNIQUELY SCRIPT EVERY ENDPOINT

SYSTEM / LEVEL / DEFINITION
- CLASSICAL CLAIM → regional climate favours a relatively stable climax
- PRESSURE → disturbance history and species arrival differ among sites

STRUCTURE-FUNCTION MECHANISM
- MODERN READING → multiple plausible stable states can occur
- DISTURBANCE → redirects or resets trajectories

SCALE / EVIDENCE LIMIT
- VERDICT → climate constrains but does not uniquely script every endpoint
- CLASSICAL CLAIM → regional climate favours a relatively stable climax

- AUTOGENIC → organisms alter litter, soil, shade or microclimate
- ALLOGENIC → flood, fire, erosion, deposition, climate or land use acts externally

- FACILITATION → early species improve later establishment
- INHIBITION OR TOLERANCE → alternative replacement mechanisms

- RULE → identify the driver before naming the mechanism
- AUTOGENIC → organisms alter litter, soil, shade or microclimate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → identify the driver before naming the mechanism.

06

STAGE 06

CLIMAX THEORY PRESSURE TEST

CLIMAX THEORY PRESSURE TEST · CLASSICAL CLAIM · REGIONAL CLIMATE FAVOURS A RELATIVELY STABLE CLIMAX · DISTURBANCE HISTORY AND SPECIES ARRIVAL DIFFER AMONG SITES · MODERN READING · MULTIPLE PLAUSIBLE STABLE STATES CAN OCCUR · REDIRECTS OR RESETS TRAJECTORIES · CLIMATE CONSTRAINS BUT DOES NOT UNIQUELY SCRIPT EVERY ENDPOINT

SYSTEM / LEVEL / DEFINITION

- CLASSICAL CLAIM → regional climate favours a relatively stable climax
- PRESSURE → disturbance history and species arrival differ among sites

STRUCTURE-FUNCTION MECHANISM

- MODERN READING → multiple plausible stable states can occur
- DISTURBANCE → redirects or resets trajectories

SCALE / EVIDENCE LIMIT

- VERDICT → climate constrains but does not uniquely script every endpoint
- CLASSICAL CLAIM → regional climate favours a relatively stable climax

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → climate constrains but does not uniquely script every endpoint.

07

STAGE 07

RESTORATION STAGING LADDER

RESTORATION STAGING LADDER · 1 STABILISE · EROSION, SUBSTRATE AND HYDROLOGY · 2 REBUILD FUNCTION · ORGANIC MATTER, SOIL BIOTA AND NUTRIENT PROCESSES · 3 ENABLE NATIVE RECRUITMENT · PROPAGULES AND SUITABLE MICROCLIMATE · 4 RECOVER STRUCTURE

1 STABILISE → erosion, substrate and hydrology

2 REBUILD FUNCTION → organic matter, soil biota and nutrient processes → 3 ENABLE NATIVE RECRUITMENT → propagules and suitable microclimate → 4 RECOVER STRUCTURE → composition, layering and connectivity

SYSTEM / LEVEL / DEFINITION

- 1 STABILISE → erosion, substrate and hydrology
- 2 REBUILD FUNCTION → organic matter, soil biota and nutrient processes

STRUCTURE-FUNCTION MECHANISM

- 3 ENABLE NATIVE RECRUITMENT → propagules and suitable microclimate
- 4 RECOVER STRUCTURE → composition, layering and connectivity

SCALE / EVIDENCE LIMIT

- 5 MONITOR → function and trajectory, not planting count alone
- 1 STABILISE → erosion, substrate and hydrology

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 5 MONITOR → function and trajectory, not planting count alone.

08

STAGE 08

SCALE VOCABULARY

SCALE VOCABULARY · PLACE USED BY A SPECIES · BOUNDED INTERACTING BIOTIC AND ABIOTIC FUNCTIONAL UNIT · BROAD REGIONAL ECOLOGICAL CATEGORY · FOREST LEGAL CATEGORY · STATUS UNDER APPLICABLE LAW OR INTERPRETATION · ONE TERM CANNOT SILENTLY SUBSTITUTE FOR ANOTHER

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
HABITAT → place used by a species	BIOME → broad regional ecological category	RULE → one term cannot silently substitute for another
ECOSYSTEM → bounded interacting biotic and abiotic functional unit	FOREST LEGAL CATEGORY → status under applicable law or interpretation	HABITAT → place used by a species

ECOLOGICAL UNIT / PARAMETER

- HABITAT → place used by a species
- ECOSYSTEM → bounded interacting biotic and abiotic functional unit

DECISIVE DISTINCTION

- BIOME → broad regional ecological category
- FOREST LEGAL CATEGORY → status under applicable law or interpretation

SCALE / STATUS / EXCEPTION

- RULE → one term cannot silently substitute for another
- HABITAT → place used by a species

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → one term cannot silently substitute for another.

09

STAGE 09

BIOME CONTROL MATRIX

BIOME CONTROL MATRIX · BROAD CLIMATE · TEMPERATURE AND PRECIPITATION PATTERN · SOIL AND TOPOGRAPHY · LOCAL WATER AND NUTRIENT CONDITIONS · FIRE AND HERBIVORY · MAINTAIN OR REDIRECT OPEN VEGETATION · LAND-USE HISTORY

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
BROAD CLIMATE → temperature and precipitation pattern	FIRE AND HERBIVORY → maintain or redirect open vegetation	BOUNDARY → biome maps are gradients, not exact site-level lines
SOIL AND TOPOGRAPHY → local water and nutrient conditions	LAND-USE HISTORY → modifies present composition	BROAD CLIMATE → temperature and precipitation pattern

ECOLOGICAL UNIT / PARAMETER

- BROAD CLIMATE → temperature and precipitation pattern
- SOIL AND TOPOGRAPHY → local water and nutrient conditions

DECISIVE DISTINCTION

- FIRE AND HERBIVORY → maintain or redirect open vegetation
- LAND-USE HISTORY → modifies present composition

SCALE / STATUS / EXCEPTION

- BOUNDARY → biome maps are gradients, not exact site-level lines
- BROAD CLIMATE → temperature and precipitation pattern

09

STAGE 09

BIOME CONTROL MATRIX

BIOME CONTROL MATRIX · BROAD CLIMATE · TEMPERATURE AND PRECIPITATION PATTERN · SOIL AND TOPOGRAPHY · LOCAL WATER AND NUTRIENT CONDITIONS · FIRE AND HERBIVORY · MAINTAIN OR REDIRECT OPEN VEGETATION · LAND-USE HISTORY

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
BROAD CLIMATE → temperature and precipitation pattern	FIRE AND HERBIVORY → maintain or redirect open vegetation	BOUNDARY → biome maps are gradients, not exact site-level lines
SOIL AND TOPOGRAPHY → local water and nutrient conditions	LAND-USE HISTORY → modifies present composition	BROAD CLIMATE → temperature and precipitation pattern

ECOLOGICAL UNIT / PARAMETER
- BROAD CLIMATE → temperature and precipitation pattern
- SOIL AND TOPOGRAPHY → local water and nutrient conditions

DECISIVE DISTINCTION
- FIRE AND HERBIVORY → maintain or redirect open vegetation
- LAND-USE HISTORY → modifies present composition

SCALE / STATUS / EXCEPTION
- BOUNDARY → biome maps are gradients, not exact site-level lines
- BROAD CLIMATE → temperature and precipitation pattern

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOUNDARY → biome maps are gradients, not exact site-level lines.

10

STAGE 10

OPEN ECOSYSTEM FIREWALL

OPEN ECOSYSTEM FIREWALL · GRASSLAND, SAVANNA, SCRUB · MAY BE NATURAL ECOLOGICAL STATES · ADMINISTRATIVE OR REVENUE LABEL · LEGAL CATEGORY IN THE RELEVANT CONTEXT · LAND-COVER INTERVENTION, NOT PROOF OF RESTORED BIOME · ASSESS NATIVE BIOME BEFORE AFFORESTATION

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
GRASSLAND, SAVANNA, SCRUB → may be natural ecological states	FOREST → legal category in the relevant context	DECISION → assess native biome before afforestation
WASTELAND → administrative or revenue label	PLANTATION → land-cover intervention, not proof of restored biome	GRASSLAND, SAVANNA, SCRUB → may be natural ecological states

ECOLOGICAL UNIT / PARAMETER
- GRASSLAND, SAVANNA, SCRUB → may be natural ecological states
- WASTELAND → administrative or revenue label

DECISIVE DISTINCTION
- FOREST → legal category in the relevant context
- PLANTATION → land-cover intervention, not proof of restored biome

SCALE / STATUS / EXCEPTION
- DECISION → assess native biome before afforestation
- GRASSLAND, SAVANNA, SCRUB → may be natural ecological states

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PLANTATION → land-cover intervention, not proof of restored biome.

11

STAGE 11

PYQ AND MONITORING BOUNDARY

PYQ AND MONITORING BOUNDARY · 2021 ROUTE · PIONEERS SURVIVING ON SURFACES WITHOUT SOIL · 2024 ESSAY · FORESTS AND DESERTS USED AS A QUALIFIED ECOLOGICAL WARNING · FOREST COVER · CANOPY TREND, NOT SUCCESSIONAL-STAGE PROOF · RESTORATION CLAIM

2021 ROUTE → pioneers surviving on surfaces without soil → 2024 ESSAY → forests and deserts used as a qualified ecological warning → FOREST COVER → canopy trend, not successional-stage proof → RESTORATION CLAIM → needs repeated composition, soil and function evidence → LIVE STUB → adds no succession timeline, biome range or status

ECOLOGICAL UNIT / PARAMETER
- 2021 ROUTE → pioneers surviving on surfaces without soil
- 2024 ESSAY → forests and deserts used as a qualified ecological warning

DECISIVE DISTINCTION
- FOREST COVER → canopy trend, not successional-stage proof
- RESTORATION CLAIM → needs repeated composition, soil and function evidence

SCALE / STATUS / EXCEPTION
- LIVE STUB → adds no succession timeline, biome range or status
- 2021 ROUTE → pioneers surviving on surfaces without soil

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIVE STUB → adds no succession timeline, biome range or status.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · FOREST SURVEY OF INDIA (FSI) · BIENNIAL STATE OF FOREST REPORTS TRACK CANOPY-DENSITY · NATIONAL AFFORESTATION PROGRAMME · CAMPA-FUNDED PROJECTS · IMPLEMENTATION GAP · AFFORESTATION TARGETS ARE FREQUENTLY REPORTED IN AREA (HECTARES · "FOREST COVER" AS MEASURED BY CANOPY-DENSITY THRESHOLD (FSI METHODOLOGY)

OPTIONAL ECOLOGICAL PROCESS DEPTH
- Forest Survey of India (FSI): biennial State of Forest Reports track canopy-density
- MoEFCC / National Afforestation Programme / CAMPA-funded projects: implement
- CAUTION: Implementation gap: afforestation targets are frequently reported in area (hectares
- CAUTION: "Forest cover" as measured by canopy-density threshold (FSI methodology) cannot by

SCALE, PARAMETER OR STATUS LIMIT
- CAUTION: Long-term successional trajectories (decades to centuries for true climax forest) are
- CAUTION: Biome boundaries are not sharp lines but gradients (ecotones, Topic 01); treating biome
- Primary succession lacks a pre-existing soil layer; secondary succession retains soil
- Hydrarch succession begins in water and trends toward a drier (mesic) climax; xerarch

COMPARATIVE RESTORATION NUANCE
- Biome classification uses temperature and precipitation as primary variables, directly
- India contains tropical wet evergreen, tropical dry deciduous, thorn/desert and montane
- NOT: Climax community composition is uniquely and permanently fixed by climate alone.
- NOT: Planting any tree species counts as ecological restoration toward climax. - Restoration

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels